

IRIS T3 Carrier Board

Modular, Hardware-Agnostic Robotics Head Platform for IRIS 2026

The **IRIS T3 Carrier Board** is a universal, low-cost, hardware-agnostic compute and sensor backbone designed for humanitarian robotics. It allows any user with an ESP32, Raspberry Pi Pico, or even a ruggedized USB-C smartphone to act as the “brain” of a T3-class robot.

This repository contains the design specifications, hardware interface definitions, and manufacturing documentation needed to produce the board across three prototype stages (v1 → v3).

Project Purpose

IRIS aims to deliver resilient, modular robotics infrastructure for emergency response, community aid, and communication restoration. The T3 Carrier Board is the foundation for:

- Decentralized sensor processing
 - Mesh network propagation (QMesh / ESP-Mesh)
 - Device-agnostic compute
 - Simulation → real-world robotics experiments
 - Low-cost field-repairable aid robots
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Board Capabilities

Core Features

- Universal MCU socket for ESP32-S3, Pico/PicoW, and compatible 3.3V microcontrollers
- USB-C phone brain option for high-level compute
- 3.3V and 5V sensor rails
- High-bandwidth connector for cameras or SPI/MIPI expansion
- Power domain isolation and robust surge protection
- Designed for rapid field repairs and modular upgrades

Compatibility

- ESP32-S3 DevKitC
- Raspberry Pi Pico / PicoW
- Raspberry Pi CM (via optional adapter)
- Any Android/iOS phone via gimbal-mounted USB-C

Board Layout

Dimensions: 85mm × 65mm

Mounting: 4 × M3 plated holes

Connectors

Location	Connector	Function
Top	USB-C	Phone compute + 5V input
Top	2 × JST-GH 4-pin	I2C sensor rails
Left	40-pin MCU header	Universal compute block
Bottom	XT30 or barrel	Power input (5–15V)
Bottom	JST-GH 6-pin	High-bandwidth sensor/camera rail
Right	Header pads	UART, SWD, debug

All connectors fully labeled on silkscreen.

Power Architecture

- **Input:** 5–15V DC
- **USB-C:** 5V VBUS input supported
- **Rails:**
 - V_HIGH: 5V
 - V_LOGIC: 3.3V
 - V_SYS: raw input
- **Regulators:**
 - Buck converter (MP1584EN footprint)
 - 3.3V LDO (AMS1117-3.3 or RT9080 footprint)

Includes polyfuse and USB ESD protection.

MCU Header Specification

A 40-pin hybrid header designed to accept multiple microcontroller form-factors.

Required signal categories:

- 2× I2C buses
- 2× SPI buses
- 2× UART buses
- At least 6 ADC channels
- Boot/Enable lines for ESP32
- 5V + 3.3V + GND distribution

Example silk labels:

```
MCU_GPIO_01
MCU_GPIO_02
MCU_UART1_TX
MCU_SPI0_MOSI
MCU_I2C0_SDA
MCU_ADC2
...
```



Sensor & Expansion Rails

I2C Rails (2× JST-GH 4-pin)

1. 3.3V
2. SDA
3. SCL
4. GND

High-Bandwidth Rail (JST-GH 6-pin)

1. 5V
2. MIPI/SPI_CLK+
3. MIPI/SPI_CLK-

- 4. DATA+
- 5. DATA-
- 6. GND

Debug Pads

- SWD
 - UART0
 - RESET + BOOT buttons
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Prototype Roadmap

Prototype 1 — Core Power + I/O

- Power regulation stable
- USB-C → MCU communication
- I2C rails validated
- Basic sensors + MCU swapping

Prototype 2 — Expansion & Phone Integration

- Gimbal-mounted phone compute
- High-bandwidth sensor rail
- Full MCU swap support
- Clean routing + isolated grounds

Prototype 3 — Field-Ready

- Reinforced connectors
- Surge and ESD-hardened
- Official mechanical mount spec
- Final BOM and gerbers ready for production

Deliverables (for Engineers)

This repository contains:

- `/schematics` → Full board schematic (PDF + KiCAD/Altium)
 - `/pcb` → PCB layout files
 - `/gerbers` → Manufacturing export set
 - `/3d-model` → STEP model for CAD/robotics integration
 - `/bom` → BOM with vendor links
 - `/docs` → Testing procedures, pinout documentation, expansion guides
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Manufacturing Notes

- 4-layer board recommended (top, GND plane, power plane, bottom).
 - Keep USB data lines impedance controlled.
 - Prefer matte-black or matte-blue solder mask for visibility.
 - Use JST-GH for all field-replaceable connectors.
 - Ground stitching vias along all board edges.
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License

All IRIS hardware is released under the **IRIS Commons Hardware License**, allowing:

- Free use for humanitarian aid, education, community research
 - Restricted use for military or extractive commercial exploitation
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Contributing

IRIS welcomes electrical engineers, firmware developers, and community makers. Forks, PRs, and discussions are open for:

- Sensor modules
 - Compute adapters
 - Simulation-to-real bridging
 - Documentation and field repair guides
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Contact

For engineering or collaboration requests: **IRIS Lab — Integrated Reality Information Systems** Email: contact pending GitHub: [RocknRye747/IRIS](https://github.com/RocknRye747/IRIS)